Actual affine Boolean contraction

Denominator clearing, integral local charts, and quotient transport

November 9, 2026. Source commit `fa4e2bbab`.

Unsigned producer evidence throughout: [pcc:unverified].

Result and assumptions. The actual source D.8 contraction admits denominator clearing with exponent exactly M, the number of real row ports. The zero-output factor is affine; its constant term must survive homogenization. For immutable $A(Y), B(X)$ and $\ell_0(z) = 1 - z$, $\ell_1(z) = z$, put

$$F(W) = \sum_{X,Y \in \{0,1\}^M} A(Y)B(X) \prod_{j=1}^{M} \ell_{Y_j}((WX)_j).$$

If D is a common unit and $n_j = (NX)_j$, each factor clears as

$$D\ell_1(n_j/D) = n_j, \qquad D\ell_0(n_j/D) = D - n_j.$$

Multiplying the M factors and distributing both sums proves

$$D^M F(N/D) = \sum_{X,Y} A(Y)B(X) \prod_j \begin{cases} n_j & Y_j = 1, \\ D - n_j & Y_j = 0. \end{cases}$$

This is valid over every commutative ring, including rings with nilpotents. The Lean proof imposes no $M > 0$ restriction.

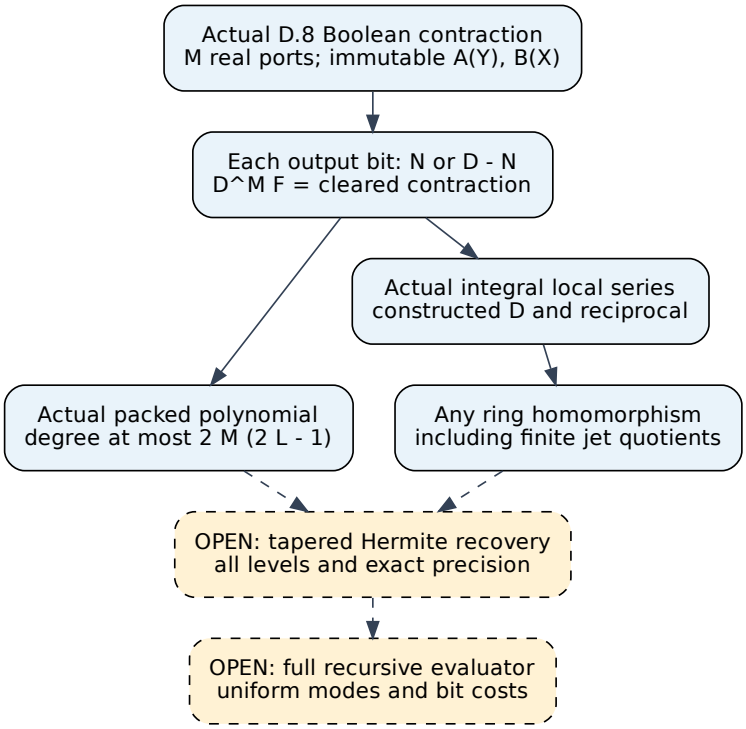
Actual packed Cauchy repair. Set $L = 2^a$, $N_{ij} = B_{ij}^2$ within a packet, and zero between packets; $D = \sum_s B_s$. The imported actual B_s has degree at most $2L - 1$. Each cleared factor therefore has degree at most $2(2L - 1)$, giving

$$\deg\big(D^M F(N/D)\big) \le 2M(2L - 1).$$

The polynomial statement is over any $\mathbb{Q}$-algebra with fixed A, B. At rational points the clearing identity requires $D(T) \neq 0$; there is no cancellation at poles. An injective supplied placement gives $U(0) = I$ and $F(U(0)) = \sum_X A(X)B(X)$. Padding has no real ports. A packing constructor and its cost are separate obligations.

Controls. $3\ell_0(2/3) = 1$, whereas $\ell_0(2) = -1$. The native identity certificate `IDC1had86c2d70ea770ef` covers the polynomial rearrangement $d(1 - zi) = d - z(di)$; the unit relation is separately proved in Lean. The unchanged native checker accepts extra evaluation points and rejects naive substitution as a false identity. This finite-field check is supporting evidence, not the quantified ring proof or an independent attestation.

Actual local series and finite quotients. The existing construction uses the concrete rational integral ring $\mathcal{O} = \{q \in \mathbb{Q} : |q|_2 \leq 1\}$. At every source node it constructs the shifted denominator series and an integral reciprocal with both inverse identities. Packaging those actual series as a unit instantiates the affine clearing theorem for the packet-restricted actual local kernel. Applying any ring homomorphism transports the entire weighted identity. This includes reduction by a finite jet ideal; arbitrary coefficient truncation is not asserted to be a ring homomorphism.



Solid boxes describe this proof component; dashed boxes remain open.

Checks. Default build `packed-build-004.log` passed. The actual repository source gate reports 140 declarations, no extra trusted axioms. Foundations are `propext`, `Classical.choice`, `Quot.sound`. The official import-inclusive fresh replay is recorded in `lean/contraction-fresh-status.json` and its raw log. Failed elaborations and the first native build are retained. No checker, trust policy, kernel or dependency revision was changed. Native module tests: 22 passed, zero failed. These are unsigned producer checks [pcc:unverified].

No timings or speedups are inferred from compiler logs. Historical comparison plots and scoreboards are unchanged because no measured value or admitted performance row changed.

Proof map. All names below are in `BooleanContraction` or `PackedContraction`. The report Markdown and `lean/contraction-claim-pins.json` bind each name and canonical statement hash to its unsigned PCC record.

Affine weighted contraction	`sourceContractionClearing`, PCC-20261109-87aebd
Actual rational point equality	`packedPointClearing`, PCC-20261109-205f61
Mapped actual polynomial degree	`actualPackedContractionDegree`, PCC-20261109-21b592
Origin under injective placement	`actualPackedOriginContraction`, PCC-20261109-7987a6
Actual integral local identity	`actualLocalContractionClearing`, PCC-20261109-9e5d39
Arbitrary ring-map transport	`mappedLocalContractionClearing`, PCC-20261109-055aa0

Sources checked. The primary manuscript is OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026, Appendix D.4.2–D.4.4, especially D.7–D.10. PDF SHA256: `5b1612dc2eebb2cad5b4cbe62d1fed2982f01ab1908c26f70010a651487e8123`. Pinned release: https://github.com/openai/math/blob/7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026/paper.pdf. Repository fallback review: common-kernel Boolean gauges, its `GaugeCounterexample.lean`, and tensor-variants matched-level review. The KB MCP is unavailable and wrapper status/search exited 127. This bounded review does not establish novelty or survey exhaustion.

Next discriminating obligation. Source D.5 uses $h = \lceil 3M/u \rceil$ and $q_t = h \max(0, u - v_2(t))$. The odd-node lower bound $Lhu \geq 3ML$ alone does not imply enough jets for doubled degree $2M(2L-1)$. Prove the total over all active levels, then construct exact Hermite recovery with its finite precision loss. The first-three-level numerical margin is useful evidence, but is not that recovery theorem. Since $D(0) = 1$, recovering the cleared value recovers the target. Local construction and powering costs must be charged.

The explicit double sum is a correctness specification with exponentially many assignments; it does not supply a faster evaluator. Uniform modes, child-entry construction and complete recursive bit costs remain open. The scalar repair still preserves the identity at the origin, so rank reduction there requires a further mechanism.

Reproduction. With pinned Lean 4.35.0 on PATH, from the repository root:

```
lake -d research/prime_common_scalar_20261108/lean build
lake -d research/prime_common_scalar_20261108/lean \
  env leanchecker --fresh --verbose PackedContraction
```

Native certificate reproduction commands and exact scope are in `identity/README.md`. Sources and all logs are versioned alongside this report.